Lecture 1. Combinatorial Analysis

Counting principle, permutations, combinations, binomial and multinomial theorems, integer solutions

Contents

1.1 The basic principle of counting **2**

1.2 Permutations **2**

1.3 Combinations **3**

1.4 Multinomial coefficients **4**

1.5 Integer solutions of equations **4**

1.6 Selected exercises **5**

Many probability questions on finite sample spaces with equally likely outcomes reduce to counting. This lecture collects the counting tools used throughout the course.

1.1 The basic principle of counting

Motivation. For equally likely outcomes, a probability is a ratio of two counts. Most early problems are therefore counting problems, and they all reduce to one rule: multiply the number of choices at each stage.

Theorem 1.1 (Generalized counting principle)

If r experiments are performed in sequence so that the first has n_1 possible outcomes and, for each outcome of the first $i-1$ experiments, the i-th has n_i possible outcomes, then the r experiments together have $n_1 n_2 \cdots n_r$ possible outcomes.

Proof. For $r = 2$ list the outcomes as pairs (i, j), $1 \le i \le n_1$, $1 \le j \le n_2$: an $n_1 \times n_2$ array. Induct on r, treating the first $r-1$ experiments as one. ∎

The number n_i may not depend on earlier outcomes, but the *set* of outcomes may.

Example 1.1 (License plates)

A plate has 3 letters followed by 4 digits. There are $26^3 \cdot 10^4 = 175{,}760{,}000$ plates. If no letter and no digit may repeat, there are $26 \cdot 25 \cdot 24 \cdot 10 \cdot 9 \cdot 8 \cdot 7 = 78{,}624{,}000$. The number of functions from an n-point set to $\{0, 1\}$ is 2^n, by the same principle.

1.2 Permutations

An ordered arrangement of n distinct objects is a **permutation**; there are $n! = n(n-1)\cdots 1$ of them, with $0! = 1$. The number of ordered selections of r objects out of n is $n(n-1)\cdots(n-r+1) = n!/(n-r)!$.

If the n objects fall into r groups of $n_1, \ldots, n_r$ mutually indistinguishable objects, each distinguishable arrangement corresponds to exactly $n_1! \cdots n_r!$ permutations of labelled objects (permute within each group), so there are

$$\frac{n!}{n_1!\, n_2! \cdots n_r!}$$

distinguishable arrangements.

Example 1.2

(a) The letters of PEPPER (three P, two E, one R) give $6!/(3!\,2!\,1!) = 60$ arrangements.
(b) Four mathematics, three chemistry, two history and one language book are shelved so that books on the same subject stand together. Order the four subject blocks (4! ways), then order within each block: $4!\,(4!\,3!\,2!\,1!) = 24 \cdot 288 = 6912$ arrangements.
(c) A class of 6 men and 4 women is ranked by an exam with no ties: $10! = 3{,}628{,}800$ rankings; if men are ranked only among men and women among women, $6!\,4! = 17{,}280$.

1.3 Combinations

The number of *unordered* groups of r objects chosen from n is

$$\binom{n}{r} = \frac{n(n-1)\cdots(n-r+1)}{r!} = \frac{n!}{(n-r)!\,r!}, \qquad 0 \le r \le n,$$

since each group is counted $r!$ times among the ordered selections. We set $\binom{n}{r} = 0$ for $r < 0$ or $r > n$. Clearly $\binom{n}{r} = \binom{n}{n-r}$.

Proposition 1.2 (Pascal's identity)

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r} \text{ for } 1 \le r \le n.$$

Proof. Fix one object. The r-subsets containing it number $\binom{n-1}{r-1}$; those not containing it number $\binom{n-1}{r}$. ∎

Theorem 1.3 (Binomial theorem)

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Proof. Expand $(x_1+y_1)\cdots(x_n+y_n)$: it is the sum of 2^n products, one for each choice of x_i or y_i from every factor. The products with exactly k of the x's are indexed by the k-subsets of $\{1,\dots,n\}$; there are $\binom{n}{k}$ of them. Now set $x_i = x$, $y_i = y$. ∎

Taking $x = y = 1$ shows that an n-set has $\sum_k \binom{n}{k} = 2^n$ subsets; $x = -1, y = 1$ gives $\sum_k (-1)^k \binom{n}{k} = 0$ for $n \ge 1$.

Combinatorial arguments ("count one set in two ways") prove many identities quickly.

Proposition 1.4

$$\text{(a)} \sum_{k=1}^{n} k\binom{n}{k} = n2^{n-1}, \qquad \text{(b)} \binom{n+m}{r} = \sum_{k=0}^{r} \binom{n}{k}\binom{m}{r-k} \quad \text{(Vandermonde)}.$$

Proof. (a) Count pairs (committee, chair) from n people. Choosing a committee of size k and then its chair gives the left side; choosing the chair first (n ways) and then any subset of the others (2^{n-1}) gives the right.

(b) Choose r people from n men and m women by first fixing the number k of men. ∎

Example 1.3 (Antennas in a row)

Of n antennas in a line, m are defective and indistinguishable, as are the $n-m$ working ones. How many orderings have no two defectives adjacent? Line up the $n-m$ working antennas; this creates $n-m+1$ gaps (including both ends), and at most one defective may go into each gap. So the answer is $\binom{n-m+1}{m}$. For instance, $n = 4, m = 2$ gives $\binom{3}{2} = 3$: DWDW, DWWD, WDWD.

1.4 Multinomial coefficients

The number of ways to divide n distinct objects into r *labelled* groups of sizes $n_1, \dots, n_r$ ($\sum n_i = n$) is

$$\binom{n}{n_1, \dots, n_r} = \binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-n_1-\cdots-n_{r-1}}{n_r} = \frac{n!}{n_1!\cdots n_r!},$$

which is the same number as in Section 1.2: assigning object j a group label is the same as arranging a word with n_i copies of label i.

Theorem 1.5 (Multinomial theorem)

$(x_1 + \cdots + x_r)^n = \sum_{n_1+\cdots+n_r=n} \binom{n}{n_1, \dots, n_r} x_1^{n_1}\cdots x_r^{n_r}$, the sum over nonnegative integer vectors.

Example 1.4 (Labelled versus unlabelled groups)

Ten children are split into an A team and a B team of five each: $\binom{10}{5,5} = 252$ ways. If the two teams are not labelled (just "two teams of five"), each division is counted twice, giving 126.

Example 1.5 (Knockout tournament)

In a knockout tournament with $n = 2^m$ players, the first round pairs the players and each pair produces one winner. The number of pairings is $n!/\{2^{n/2}(n/2)!\}$ (arrange in a row, pair neighbours, then forget the order within and among pairs), and each pairing has $2^{n/2}$ outcomes, so the first round has $n!/(n/2)!$ possible outcomes. Repeating round by round, the whole tournament has

$$\frac{n!}{(n/2)!}\cdot\frac{(n/2)!}{(n/4)!}\cdots\frac{2!}{1!} = n!$$

outcomes.

1.5 Integer solutions of equations

Motivation. Distributing identical objects into distinct boxes is a counting problem that appears in occupancy and urn models. The stars-and-bars argument turns it into choosing positions for separators.

Proposition 1.6 (Stars and bars)

There are $\binom{n-1}{r-1}$ vectors of *positive* integers with $x_1 + \cdots + x_r = n$, and $\binom{n+r-1}{r-1}$ vectors of *nonnegative* integers.

Proof. Write n stars in a row; a positive solution corresponds to choosing $r-1$ of the $n-1$ gaps between stars as dividers. For nonnegative solutions put $y_i = x_i + 1$: positive solutions of $\sum y_i = n + r$, of which there are $\binom{n+r-1}{r-1}$. ∎

Equivalently, $\binom{n+r-1}{r-1}$ counts the ways to put n indistinguishable balls into r distinguishable urns, and it is the number of terms in the multinomial expansion of $(x_1 + \cdots + x_r)^n$.

Example 1.6 (Investment)

\$20,000 is invested in units of \$1000 among 4 options. If all of it must be invested, there are $\binom{23}{3} = 1771$ strategies. If not all need be invested, add a fifth "cash" variable: $\binom{24}{4} = 10{,}626$. If each option must receive at least one unit, $\binom{19}{3} = 969$.

Example 1.7 (Antennas revisited)

In Example 1.3, an arrangement is determined by the numbers $x_1, \dots, x_{m+1}$ of working antennas before the first defective, between consecutive defectives, and after the last. These satisfy $\sum x_i = n - m$ with $x_1, x_{m+1} \geq 0$ and $x_2, \dots, x_m \geq 1$. Substituting $y_1 = x_1 + 1$, $y_{m+1} = x_{m+1} + 1$ gives a positive solution of $\sum y_i = n - m + 2$, so the count is $\binom{n-m+1}{m}$ again. Requiring at least two working antennas between defectives is now easy: substitute $y_i = x_i - 1$ for the interior terms.

1.6 Selected exercises

Motivation. These exercises are Ross's theoretical exercises for Chapter 1 (numbered as in the book). They train the main technique of the lecture, counting one set in two ways, and prove the multinomial theorem, which was stated above without proof.

Exercise 1.11 (Classifying subsets by their largest element)

Show that for $1 \leq k \leq n$,

$$\sum_{i=k}^{n} \binom{i-1}{k-1} = \binom{n}{k}.$$

Solution.

Step 1. Let $\mathcal{A}$ be the family of k-element subsets of $\{1, \dots, n\}$, so $|\mathcal{A}| = \binom{n}{k}$. Every $A \in \mathcal{A}$ has a unique largest element $\max A$, and $\max A \geq k$ because A contains k distinct positive integers. Hence $\mathcal{A}$ is the disjoint union of the families $\mathcal{A}_i = \{A \in \mathcal{A} : \max A = i\}$, $i = k, \dots, n$.

Step 2. A set $A \in \mathcal{A}_i$ is determined by its other $k-1$ elements, which form an arbitrary $(k-1)$-subset of $\{1, \dots, i-1\}$; conversely every such subset together with i lies in $\mathcal{A}_i$. So $|\mathcal{A}_i| = \binom{i-1}{k-1}$.

Step 3. Adding over i gives $\binom{n}{k} = \sum_{i=k}^{n} |\mathcal{A}_i| = \sum_{i=k}^{n} \binom{i-1}{k-1}$. ∎

Exercise 1.12 (Committees with a chair and a secretary)

Show that for $n \geq 1$,

$$\sum_{k=1}^{n} k^2 \binom{n}{k} = 2^{n-2} n(n+1).$$

(The identity $\sum_k k\binom{n}{k} = n2^{n-1}$ is Proposition 1.4(a).)

Solution. From n people we count the triples (C, a, b) where C is a nonempty committee, $a \in C$ is its chair and $b \in C$ its secretary, $a = b$ being allowed.

Step 1. (Committee first.) There are $\binom{n}{k}$ committees of size k, and for each of them k choices of a and k of b. So the number of triples is $\sum_{k=1}^{n} k^2 \binom{n}{k}$.

Step 2. (Officers first, $a = b$.) Choose the person $a = b$ in n ways; each of the other $n - 1$ people is in or out of C freely. This gives $n2^{n-1}$ triples.

Step 3. (Officers first, $a \neq b$.) Choose the ordered pair (a, b) in $n(n-1)$ ways and the rest of C among the other $n-2$ people in 2^{n-2} ways, giving $n(n-1)2^{n-2}$ triples (this is 0 when $n = 1$).

Step 4. Steps 2 and 3 count disjoint classes whose union is the set counted in Step 1, so

$$\sum_{k=1}^{n} k^2 \binom{n}{k} = n2^{n-1} + n(n-1)2^{n-2} = 2^{n-2}(2n + n^2 - n) = 2^{n-2}n(n+1).$$

■

Exercise 1.16 (Tournaments with ties)

In a tournament of n contestants the outcome is an ordering of the players with ties allowed: the players are divided into groups, the first group being those tied for first place, the next group those tied for the next place, and so on. Let $N(n)$ be the number of outcomes, with $N(0) = 1$.
(a) Show that $N(n) = \sum_{i=1}^{n} \binom{n}{i} N(n-i)$ for $n \geq 1$.
(b) Show that this is equivalent to $N(n) = \sum_{i=0}^{n-1} \binom{n}{i} N(i)$.
(c) Find $N(3)$ and $N(4)$.

Solution.

(a) Classify the outcomes by the number $i \in \{1, \ldots, n\}$ of players tied for last place. An outcome with i players last is determined by the set L of those players and by the outcome among the remaining $n-i$ players (which is an arbitrary ordering with ties of them; if $i = n$ there is exactly one, matching $N(0) = 1$). Conversely any set L of size i and any outcome of the others, with L appended as the last group, is an outcome with exactly i players last. This bijection gives $\binom{n}{i} N(n-i)$ outcomes with i players last, and summing over i proves the claim.

(b) Substitute $j = n - i$, which runs over $0, \ldots, n-1$, and use $\binom{n}{n-j} = \binom{n}{j}$: $\sum_{i=1}^{n} \binom{n}{i} N(n-i) = \sum_{j=0}^{n-1} \binom{n}{j} N(j)$.

(c) From (b), $N(1) = N(0) = 1$, $N(2) = N(0) + 2N(1) = 3$,

$$N(3) = N(0) + 3N(1) + 3N(2) = 1 + 3 + 9 = 13,$$
$$N(4) = N(0) + 4N(1) + 6N(2) + 4N(3) = 1 + 4 + 18 + 52 = 75.$$

■

Exercise 1.19 (The multinomial theorem)

Prove Theorem 1.5.

Solution.

Step 1. Write $(x_1 + \cdots + x_r)^n$ as a product of n factors and expand by the distributive law. The result is the sum, over all words $w = (i_1, \ldots, i_n) \in \{1, \ldots, r\}^n$, of the products $x_{i_1} x_{i_2} \cdots x_{i_n}$: the word records which summand is taken from each factor, and different words are different terms of the expansion.

Step 2. If n_j is the number of positions t with $i_t = j$, then by commutativity $x_{i_1} \cdots x_{i_n} = x_1^{n_1} \cdots x_r^{n_r}$, and $(n_1, \ldots, n_r)$ is a vector of nonnegative integers with sum n. Grouping the words by this vector,

$$(x_1 + \cdots + x_r)^n = \sum_{n_1 + \cdots + n_r = n} c(n_1, \ldots, n_r)\, x_1^{n_1} \cdots x_r^{n_r},$$

where $c(n_1, \ldots, n_r)$ is the number of words of length n containing exactly n_j copies of the letter j for each j.

Step 3. Such a word is an arrangement of n letters, n_j of which are alike of kind j; by Section 1.2 there are $n!/(n_1! \cdots n_r!) = \binom{n}{n_1,\ldots,n_r}$ of them. Hence $c(n_1, \ldots, n_r) = \binom{n}{n_1,\ldots,n_r}$, which is the theorem. ■